



Nonlinear Mixed-Type PDE Exhibiting Forward- and Backward-Parabolic Behavior

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ABSTRACT

- We study the nonlinear PDE

$$u u_x - u_{yy} = 0$$

on a rectangle. This equation moves information left/right in x and smooths it up/down in y .

- When $u > 0$ the signal travels to the right; when $u < 0$ it travels to the left. So different regions can move in opposite directions, giving mixed forward/backward-parabolic behavior.
- The equation is difficult to solve exactly, so we add a small extra smoothing term $-\varepsilon u_{xx}$ and solve this regularized problem on a grid.
- A finite-difference scheme plus a simple iterative solver in MATLAB show that the solution settles into two flat states $u \approx 1$ and $u \approx -1$ separated by a thin internal layer.

MODEL AND DATA

Regularized model

$$u u_x - u_{yy} - \varepsilon u_{xx} = 0, \quad 0 < \varepsilon \ll 1.$$

- $u u_x$: nonlinear convection in x ; the sign of u sets the direction.
- $-u_{yy}$: diffusion only in y ; it smooths values along vertical lines.
- $-\varepsilon u_{xx}$: very small diffusion in x (vanishing viscosity) to stabilize the computation.

Domain and boundary conditions

- Domain: $[0, 6] \times [0, 5]$.
- Left: $u(0, y) = +1$ (“high” state).
- Right: $u(6, y) = -1$ (“low” state).
- Top/bottom: no-flux in y ,

$$\partial_y u(x, 0) = \partial_y u(x, 5) = 0.$$

Physical picture

- Imagine a long tank: the left side is kept at level $+1$ and the right side at level -1 .
- Inside the tank, values can spread up and down, but they are pushed left or right depending on whether u is positive or negative.
- Where u changes sign, left-moving and right-moving regions meet, so an internal layer can appear and stay in place.

DISCRETIZATION

Finite-difference grid

- Uniform grid: $x_i = ih, i = 0, \dots, N_x, y_j = jp, j = 0, \dots, N_y$.
- Unknowns: $U_{i,j} \approx u(x_i, y_j)$.

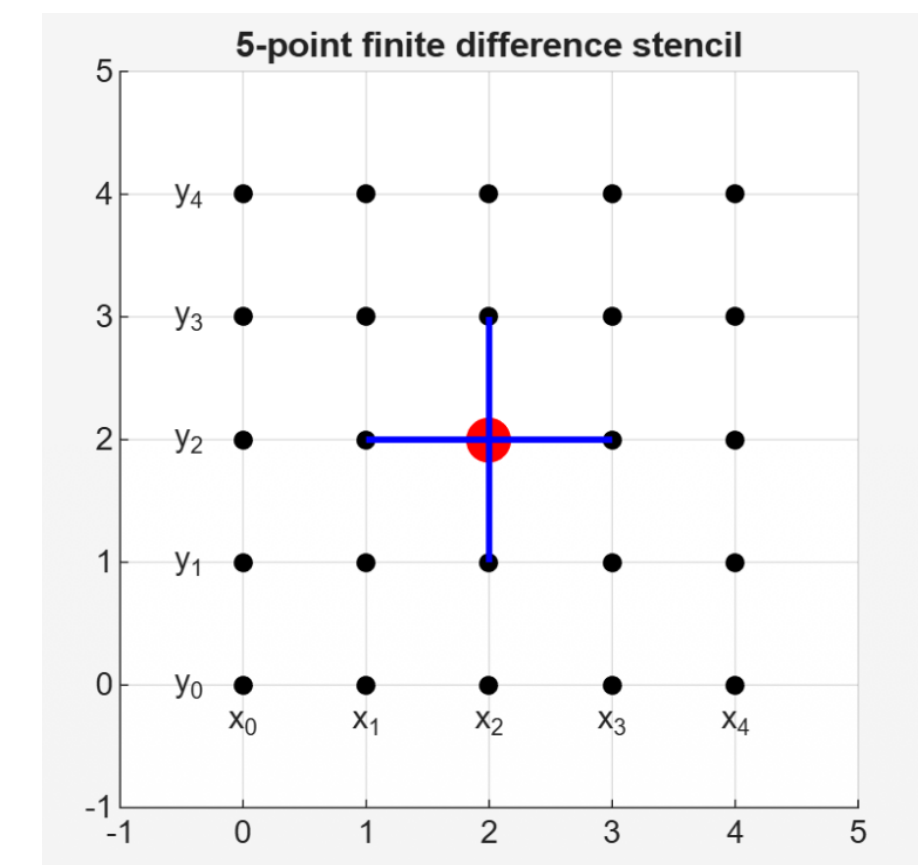
5-point finite-difference stencil

At an interior point (blue dot), the second derivative in x is approximated by

$$u_{xx}|_{\text{center}} \approx \frac{u_{\text{right}} - 2u_{\text{center}} + u_{\text{left}}}{h^2},$$

and similarly in y . This leads to a discrete equation

$$U_{i,j} U_{x,i,j} - (U_{yy})_{i,j} - \varepsilon (U_{xx})_{i,j} \approx 0.$$



LINEAR WARM-UP: INITIAL GUESS

Before tackling the nonlinear term, we solve the linear elliptic problem

$$\varepsilon u_{xx} + u_{yy} = 0$$

on the same grid and with the same boundary conditions.

- We discretize this equation with the 5-point stencil, which gives a large sparse linear system.
- MATLAB's sparse solver returns a smooth ramp from $+1$ (left) to -1 (right).
- This ramp is our initial guess $U^{(0)}$ for the nonlinear solve.

NONLINEAR SOLVER (HIGH-LEVEL VIEW)

We seek a discrete steady state of

$$u u_x - u_{yy} - \varepsilon u_{xx} = 0$$

using an iterative scheme.

- Start from $U^{(0)}$ (linear ramp).
- For iterate k , treat $U^{(k)}$ as fixed in the nonlinear term:

$$U^{(k)} U_x - U_{yy} - \varepsilon U_{xx} \approx 0.$$

- This gives a linear system for $U^{(k+1)}$ with the same boundary data.
- Repeat until the relative change is small:

$$\frac{\|U^{(k+1)} - U^{(k)}\|}{\|U^{(k)}\|} < 10^{-3}.$$

Practical choices

- Grid: $N_x = 320, N_y = 270$; spacings h, p chosen so $h, p < 2\varepsilon$.
- ε decreased from 0.1 to 0.01 in several steps.
- At each ε we allow up to 30 nonlinear iterations, usually converging sooner.

ITERATION SNAPSHOTS

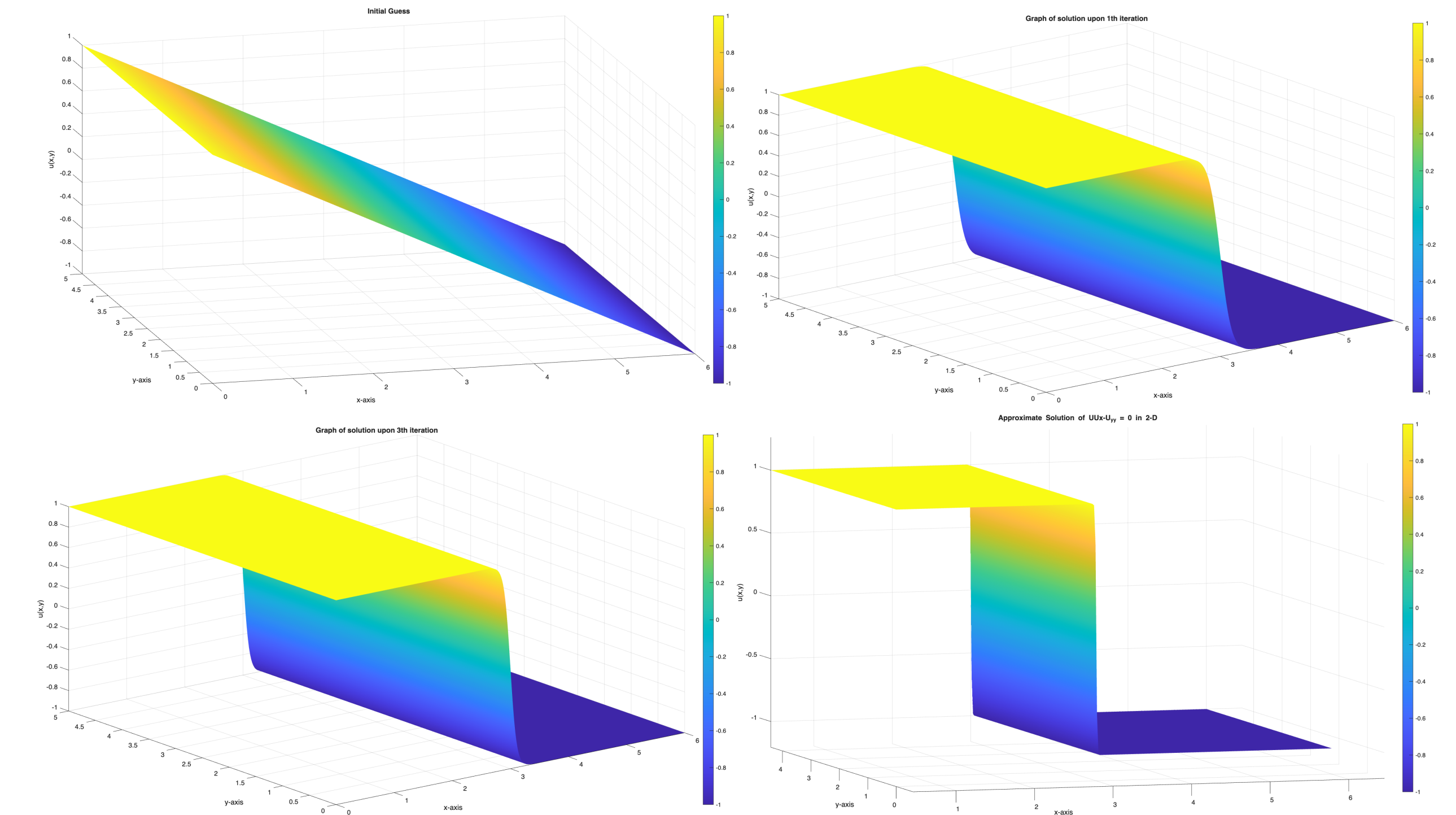


Figure 1: *

Top left (PDE1): linear initial guess. Top right (PDE2): after 1st nonlinear iteration. Bottom left (PDE3): after 3rd iteration. Bottom right (PDE4): final stabilized profile approximating $u u_x - u_{yy} = 0$.

RESULTS & INTERPRETATION

From ramp to layer

- Initial guess (PDE1).** Almost a straight ramp: diffusion in both directions dominates; no sharp structures.
- After 1st iteration (PDE2).** The ramp bends and a transition region near $x \approx 3$ appears.
- After 3rd iteration (PDE3).** A clear vertical internal layer forms. The “yellow” plateau ($u \approx 1$) and “blue” plateau ($u \approx -1$) are separated by a narrow zone.
- Final profile (PDE4).** Further iterations barely change the plot, indicating convergence for the chosen ε sequence.

Effect of the parameter ε

- Smaller ε makes the internal layer thinner but keeps its location almost fixed.
- This behavior is consistent with the vanishing-viscosity picture: as $\varepsilon \rightarrow 0$, the numerical solution approaches a sharp interface separating two steady states.

CONCLUSION

A basic finite-difference grid and an iterative linearization are enough to see a mixed-type PDE in action: a smooth ramp (PDE1) evolves into a sharp internal layer (PDE3) and stabilizes (PDE4). This pictures how nonlinear transport and anisotropic diffusion interact in a simple, concrete example.

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Thank you, Tian Jing, for supporting our research project.